

Whitelist Manager

Splunk Admin Installation Guide

Before / During / After Installation

Version 1.0.0 | February 2026

Prepared for: Splunk Infrastructure Team

BEFORE Installation (Pre-flight Checklist)

1. Verify Splunk Version

The app requires Splunk Enterprise 8.x or 9.x with Python 3 support. It has been tested on Splunk 9.3.1.

```
$SPLUNK_HOME/bin/splunk version
```

Confirm the output shows 8.0+ or 9.x.

2. Verify Python 3 Is Enabled

The app uses `python.version = python3`. Check the current setting:

```
$SPLUNK_HOME/bin/splunk btool server list --debug | grep python.version
```

If the system-level setting forces Python 2, Python 3 must be enabled. In most Splunk 8.x+ installations, Python 3 is the default.

3. Check for Naming Conflicts

The app creates the following objects. Verify none already exist:

Object	Type	Check Command
wl_manager	App	splunk display app
wl_audit	Index	splunk list index
wl_editor	Role	splunk list role
wl_viewer	Role	splunk list role
/custom/wl_manager	REST endpoint	(unique per app)

If any of these already exist, coordinate with the Security Engineering team before proceeding.

4. Review Disk Space for wl_audit Index

The app creates a `wl_audit` index with these defaults:

Setting	Value	Meaning
maxTotalDataSizeMB	1024	Maximum 1 GB of indexed data
frozenTimePeriodInSecs	94608000	Data retained for 3 years

Verify at least 2 GB of free disk space at `$SPLUNK_DB` (1 GB for data + overhead).

If your organization requires custom index paths or different retention, you can create a `local/indexes.conf` override after installation.

5. Review Network Requirements

The app makes an internal HTTPS call from the REST handler to Splunk's own management port:

```
https://127.0.0.1:8089/services/receivers/simple
```

This is a localhost-only call (Python handler to same Splunk instance) used to index audit events. No firewall changes are needed. However, if your Splunk uses a custom management port or custom SSL certificates, see the 'Special

Considerations' section.

6. Prepare RBAC Role Assignments

The app creates two new roles:

Role	Inherits	Capabilities
wl_editor	user	Read + write whitelists, access wl_audit index
wl_viewer	user	Read-only access to whitelists and wl_audit

Prepare a list of users who need each role. You will assign these after installation.

7. Identify app_context Folder Names

The master mapping CSV references CSV files in other Splunk apps. The app_context value must exactly match the app's folder name on disk.

```
ls $SPLUNK_HOME/etc/apps/ | grep -i -E "security|SA-|DA-"
```

Share these folder names with the Security Engineering team.

8. Backup (Recommended)

```
# Backup apps
tar -czf /tmp/splunk_apps_backup_$(date +%Y%m%d).tar.gz \
  $SPLUNK_HOME/etc/apps/

# Backup roles and users
$SPLUNK_HOME/bin/splunk list role > /tmp/roles_backup.txt
$SPLUNK_HOME/bin/splunk list user > /tmp/users_backup.txt
```

DURING Installation

Step 1: Install the .spl Package

Option A - Splunk Web (recommended)

- Log in to Splunk Web as admin
- Navigate to Apps > Manage Apps
- Click 'Install app from file'
- Browse to wl_manager-1.0.0.spl and click Upload
- Check 'Restart Splunk' when prompted

Option B - Splunk CLI

```
$SPLUNK_HOME/bin/splunk install app /path/to/wl_manager-1.0.0.spl \  
-auth admin:password  
$SPLUNK_HOME/bin/splunk restart
```

Option C - Manual (clustered / restricted environments)

```
cd $SPLUNK_HOME/etc/apps/  
tar -xzf /path/to/wl_manager-1.0.0.spl  
chown -R splunk:splunk wl_manager/  
$SPLUNK_HOME/bin/splunk restart
```

Step 2: Verify No Errors During Startup

After Splunk restarts, check logs for errors:

```
grep -i "wl_manager\|WhitelistHandler" \  
$SPLUNK_HOME/var/log/splunk/splunkd.log | tail -20  
  
grep -i "wl_handler\|wl_manager" \  
$SPLUNK_HOME/var/log/splunk/python_stderr.log | tail -20
```

Expected: No errors. If you see ImportError, check Python 3 configuration.

Step 3: Verify the REST Endpoint

```
curl -sk -u admin:password \  
'https://localhost:8089/services/custom/wl_manager?action=get_mapping'
```

Expected: JSON response with mapping data. If 404, restart Splunk one more time.

AFTER Installation (Post-installation Setup)

1. Verify the wl_audit Index Exists

```
$SPLUNK_HOME/bin/splunk list index wl_audit
```

Or in Splunk Web: Settings > Indexes > find wl_audit.

If the index does not appear, create it manually:

```
$SPLUNK_HOME/bin/splunk add index wl_audit \
  -maxTotalDataSizeMB 1024 \
  -frozenTimePeriodInSecs 94608000
```

2. Verify the App Is Visible

- Open Splunk Web
- Click the Apps dropdown in the top navigation
- 'Whitelist Manager' should appear in the list
- Click it - the main dashboard should load with two dropdowns

3. Assign Roles to Users

For analysts who need to EDIT whitelists:

```
$SPLUNK_HOME/bin/splunk edit user <username> -role wl_editor -auth admin:password
```

For read-only users:

```
$SPLUNK_HOME/bin/splunk edit user <username> -role wl_viewer -auth admin:password
```

Or via Splunk Web: Settings > Access Controls > Users > [user] > Edit > Roles

4. Populate the Master Mapping CSV

NOTE: This is the most critical post-installation step.

The Security Engineering team will provide the mapping data. The admin may need to assist with verifying that app_context values match actual folder names.

Edit via Splunk Web: Settings > Lookups > Lookup table files > rule_csv_map

Or directly on the file system:

```
vi $SPLUNK_HOME/etc/apps/wl_manager/lookups/rule_csv_map.csv
```

Format (one row per rule-to-CSV relationship):

```
rule_name, csv_file, app_context
DR20_malicious_command, DR20_whitelist.csv, SplunkEnterpriseSecuritySuite
DR45_suspicious_login, DR45_whitelist_users.csv, SplunkEnterpriseSecuritySuite
DR45_suspicious_login, DR45_whitelist_hosts.csv, SplunkEnterpriseSecuritySuite
```

5. Verify CSV File Permissions

The Splunk process (running as the 'splunk' user) needs read and write access to every CSV file referenced in the mapping.

```
# Check permissions on referenced CSV files
ls -la $SPLUNK_HOME/etc/apps/<app_context>/lookups/<csv_file>
```

Fix permissions if needed:

```
chown splunk:splunk $SPLUNK_HOME/etc/apps/<app>/lookups/<file>.csv
chmod 644 $SPLUNK_HOME/etc/apps/<app>/lookups/<file>.csv
```

6. Test End-to-End

- Log in as a user with the wl_editor role
- Open Apps > Whitelist Manager
- Select a detection rule, then a CSV file
- The table should load with CSV contents
- Modify a cell, enter a comment, click Save
- Go to the Audit Trail tab - the change should appear

Verify in SPL:

```
index=wl_audit sourcetype=wl_audit | head 5
| spath
| table timestamp analyst detection_rule csv_file comment
      rows_added rows_removed
```

7. Test RBAC Enforcement

- Log in as a user WITHOUT the wl_editor role
- Open the Whitelist Manager dashboard
- Try to save a change - should show 'Permission denied'

Special Considerations

Search Head Cluster (SHC) Deployment

- Place the app in: \$SPLUNK_HOME/etc/shcluster/apps/wl_manager/
- Push the bundle: splunk apply shcluster-bundle -target <captain_uri>
- The wl_audit index must also be configured on the indexers

Indexer Cluster Deployment

The wl_audit index must be created on indexers via the cluster master:

```
mkdir -p $SPLUNK_HOME/etc/master-apps/wl_manager/default/  
cp indexes.conf $SPLUNK_HOME/etc/master-apps/wl_manager/default/  
$SPLUNK_HOME/bin/splunk apply cluster-bundle
```

Custom Management Port or SSL

If your Splunk uses a non-default management port (not 8089), coordinate with the Security Engineering team to update the handler's _index_audit() method.

Monitoring (Optional)

Consider a saved search to alert on large whitelist changes:

```
index=wl_audit sourcetype=wl_audit  
| spath  
| where rows_added > 10 OR rows_removed > 10  
| table timestamp analyst detection_rule csv_file  
      rows_added rows_removed comment
```

Quick Reference Card

Item	Value
App folder	\$SPLUNK_HOME/etc/apps/wl_manager/
REST endpoint	https://<splunk>:8089/services/custom/wl_manager
Web dashboard	https://<splunk>:8000/app/wl_manager/whitelist_manager
Audit index	wl_audit
Audit log file	\$SPLUNK_HOME/var/log/splunk/wl_manager_audit.log
Mapping CSV	\$SPLUNK_HOME/etc/apps/wl_manager/lookups/rule_csv_map.csv
Edit role	wl_editor
View role	wl_viewer
Python version	Python 3
Splunk version	8.x+ / 9.x (tested on 9.3.1)

Uninstallation

To remove the app (preserves audit data):

```
$SPLUNK_HOME/bin/splunk remove app wl_manager -auth admin:password  
$SPLUNK_HOME/bin/splunk restart
```

To also remove the audit index and its data:

```
$SPLUNK_HOME/bin/splunk remove index wl_audit
```

The custom roles (wl_editor, wl_viewer) are removed with the app. Users who had these roles will lose them automatically.